

No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. All answers should be fully simplified, but may contain factorials, powers, binomial coefficients and Stirling numbers in instances where those cannot easily be evaluated. Write your date of birth on each sheet of paper.

1. Find all positive integers $x \leq 1000$ satisfying the system of congruencies

$$\begin{cases} x \equiv 1 \pmod{2} \\ x \equiv 0 \pmod{3} \\ x \equiv 1 \pmod{5} \\ x \equiv 2 \pmod{7} \end{cases}$$

2. Solve the recurrence relation $a_{n+2} + 4a_{n+1} + 3a_n = 16n^2 - 10$ with initial conditions $a_0 = -8$ and $a_1 = 21$.
3. We have a bag full of marbles in the colours red, blue, green, black and white. In how many ways can we select 10 marbles? (We care only about the resulting collection of marbles, not the order in which they are picked up.)
- a) If there are 20 marbles of each colour in the bag?
 - b) If there are five red, five blue, five green, four black and four white marbles in the bag?
 - c) If the number of marbles of each colour is as in **b**, but we may only make selections with an even number of black marbles and an even number of white marbles?
4. Look at loop-free undirected graphs $G = (V, E)$ with five vertices. For each possible value of $|E|$ = the number of edges, find out if
- (i) all such graphs contain a Hamilton cycle or
 - (ii) some but not all such graphs contain a Hamilton cycle or
 - (iii) no such graphs contain a Hamilton cycle.

Remember to prove your statement for each $|E|$.

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5. Consider the linear code C over $\mathbb{Z}_{11}$ with control matrix

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \end{pmatrix}$$

- a) How many words does C contain?
- b) Find the separation of C .
- c) Find a pair of words in C at minimal distance.
- d) For each of the words $w_1 = (3 \ 6 \ 1 \ 1 \ 0)$, $w_2 = (4 \ 10 \ 10 \ 1 \ 0)$ and $w_3 = (5 \ 1 \ 1 \ 1 \ 1)$ check if it is a code word. If not correct the word if possible. (Remember to explain why it can or cannot be corrected.)

6. Let $R = \{a + b\omega \mid a, b \in \mathbb{Z}\}$ where $\omega = e^{\frac{2\pi i}{3}}$.

- a) Show that R , equipped with ordinary addition and multiplication of complex numbers, is a ring with unity. Does R have zero divisors?
- b) Show that if $r = a + b\omega \in R$ then $|r|^2$ is always an integer. (Here $|\cdot|$ denotes the usual modulus for complex numbers.)
- c) Find all units in R . *Hint:* Use the result from **b**.
- d) Is R a field?